

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and trained **deep learning networks** for object classification and scene understanding, achieving approximately **0.88 AUC** on a held-out dataset of vehicle trajectories.
- Selected and engineered appropriate **datasets** and data representation methods for training complex neural networks, ensuring data integrity for downstream model performance.
- Collaborated with data engineers to construct robust **data pipelines** that supported the acquisition and processing of large-scale automotive sensor data, reducing data latency by **18%**.
- Performed statistical analysis and fine-tuning of model hyperparameters based on test results, iterating on network architectures to improve accuracy by **7%**.
- Wrote Python scripts to automate data validation and prepare datasets for **machine learning frameworks** like PyTorch, processing over **500GB** of raw sensor logs weekly.
- Contributed to the development of a Python-based simulation environment for testing autonomous driving algorithms, integrating **computer vision** modules for environmental perception.

SKILLS

Machine Learning – Deep Learning, Neural Networks, Keras, PyTorch, scikit-learn, Model Training, Inference, Statistical Analysis, Hyperparameter Tuning

Computer Vision – 3D Object Detection, Pedestrian Classification, Vehicle Classification, Trajectory Mapping, Online Calibration, Lens Soiling Detection, Collision Mitigation, Lane Detection, Park-Slot Detection, OpenCV

Programming & Data – Python, SQL, Data Structures, Data Modeling, Data Pipelines, ETL, Pandas, NumPy, R

Development Tools & Platforms – Linux, Git, Jenkins, Docker, AWS (EC2, S3, Lambda, CloudWatch)

Automotive & Mobility – Automated Driving Systems, Sensor Fusion, Embedded Platforms, Imaging Systems

PROJECTS

Deep Learning for Environmental Perception

- Engineered a **convolutional neural network (CNN)** architecture in PyTorch for classifying road objects and markings from simulated automotive sensor data, achieving **0.91 accuracy** on a 150k-image dataset.
- Developed Python scripts to preprocess and augment image datasets, implementing data augmentation techniques that improved model generalization by **6%**.
- Integrated the trained model into a Python-based simulation framework, enabling real-time object detection and contributing to a **15%** reduction in simulation setup time.

Automated Data Pipeline for Sensor Fusion

- Designed and implemented a Python data pipeline on AWS, processing and harmonizing data from multiple simulated sensor streams (camera, LiDAR) for downstream **sensor fusion** algorithms.
- Automated data ingestion and validation steps using **Pandas** and SQL, reducing manual data preparation effort by approximately **10 hours** per week.
- Containerized the data pipeline using **Docker** for consistent deployment across development and testing environments.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)